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**ADAPT: AI-DRIVEN CUSTOMIZABLE CHATBOT PLUGIN FOR
ENHANCED USER INTERACTION IN WEB-BASED PLATFORMS**

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ABSTRACT

This study presents the development of ADAPT: AI-Driven Customizable Chatbot Plugin for Enhanced User Interaction in Web-Based Platforms, designed to provide a versatile and scalable solution for sectors such as academia, retail, and healthcare. The chatbot leverages advanced natural language processing (NLP) algorithms and machine learning techniques to deliver effective user interaction and accurate customer assistance. It ensures seamless multi-platform compatibility, offering real-time response capabilities and reliability for uninterrupted service. Additionally, the system is designed for secure integration with existing platforms and emphasizes maintainability to address both functional and non-functional requirements. By focusing on customization and adaptability, this research highlights the potential of AI-driven chatbot solutions to enhance user engagement and operational efficiency across various industries.

INTRODUCTION

Background and Rationale of the Study

Communication systems have become a critical component of organizations, as they rely on these systems to effectively transmit information to customers, ensuring that the exchange of services aligns accurately with the customer's needs as part of delivering quality customer service (Zang, 2023). In today's highly competitive business environment, customer service plays a crucial role in the success of any organization. With the constant growth of e-commerce and the increasing need for immediate feedback, providing users with a satisfying and high-quality experience has become essential (Valderrama & Aguilar-Alonso, 2023).

One technology that has become integral to enhancing customer service is the chatbot. Chatbots are now found on various platforms, including websites, messaging applications, and standalone messaging systems. A chatbot is a computer program that simulates human conversation through text, allowing it to understand users' questions and automate responses (Haristiani, 2019). This is often accomplished using Natural Language Processing (NLP), a branch of artificial intelligence that enables computers to comprehend and generate human language (Chen, 2021).

Chatbots are transforming the way businesses operate by improving the customer experience in several crucial ways. They offer 24/7 availability, provide instant responses, and

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	ensure consistent answers, which significantly enhances customer satisfaction. On the business side, chatbots reduce operational costs, serve more customers simultaneously, and boost customer engagement by maintaining constant interaction (Maderis, 2024). As Gallemard (2022) points out, customers prefer to receive immediate responses to their inquiries, which can directly influence their decision to proceed with a business's services or explore alternatives. Without chatbots, repetitive tasks like answering frequently asked questions accumulate, reducing the efficiency of handling large volumes of inquiries at scale. Many existing chatbot solutions face limitations, such as restricted customization, platform compatibility issues, and the lack of advanced AI features like nuanced language understanding and real-time adaptability. Other gaps include poor scalability, maintainability, and high implementation costs. These challenges highlight the need for a more advanced and customizable chatbot plugin that addresses these shortcomings effectively. Given these challenges, the study titled "ADAPT: AI-Driven Customizable Chabot Plugin for Enhanced User Interaction in Web-Based Platforms" seeks to introduce a customizable chatbot platform across various sectors, including academic, retail, and medical industries. This platform will leverage NLP algorithms and machine learning techniques to ensure effective user interaction and accurate customer assistance. Additionally, the chatbot will ensure seamless multi-platform compatibility, real-time response capabilities, and reliability for uninterrupted service. The study also focuses on maintainability and secure integration with existing systems, ensuring the platform meets both functional and non-functional requirements. Objectives of the Study	
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	1. To design and develop a robust AI-driven chatbot plugin that utilizes advanced natural language processing and machine learning techniques to accurately understand and generate responses to user queries across multiple web-based platforms. 2. To incorporate a suite of customizable features within the chatbot plugin, allowing users to adjust dialogue flows, response templates, and integration settings to suit their specific needs and preferences. 3. To integrate the chatbot plugin seamlessly with multiple popular web-based platforms, ensuring compatibility and effective performance within diverse web environments. 4. To assess the effectiveness of the chatbot plugin through rigorous testing, including user feedback, engagement metrics, and system performance analysis, to verify improvements in user interaction and satisfaction across the supported platforms. Significance of the Study The study aims to benefit to the following: End Users. This study will help end users understand how using chatbots may improve their digital platforms experience by offering personalized service, fast resolution to issues, and quick responses to queries. Businesses. Organizations and departments can benefit significantly from the findings of this study on chatbots by gaining a deeper understanding of how to effectively implement and leverage these technologies.	
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Web Developers. This study will provide detailed analysis and recommendations on integrating these programs effectively into web projects, addressing common configuration challenges and enhancing overall performance.

Future Researchers. This study brings out the possibilities of configurable chatbots for user interaction on digital platforms and offers insights into their implementation and its effectiveness. It serves to open possibilities for more study that focusses on examining advanced artificial intelligence (AI) technologies and enhancing chatbot capabilities.

Scope and Limitation

Advanced Dialogue and Adaptive Plugin Tool (ADAPT) a customizable chatbot plugin tailored for web development environment, but the scope of the testing phase will span on multiple industries, including academic, retail, and healthcare. The goal of “ADAPT” is to provide a solution for addressing diverse user needs, including customers, employees, and stakeholders. The project will be developed, tested, and deployed within the academic school year of 2024-2025. It will integrate intent classification and natural language processing models. It will have an interface for non-technical users for customizing the chatbot. The system will run on both mobile and desktop devices. For an optimal user experience on the administration settings in configuring the chatbot, it is best to use desktop devices. For mobile view, the configuration interface will only offer basic functionalities.

“ADAPT” will avoid handling sensitive or confidential information. The chatbot will only be limited and respond in English language, the chatbot will only rely on, and answer questions

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter provides an overview of related literature that explains and describes the key ideas and information discussed in the study, using data and information gathered from books, peer-reviewed journals, and online articles.

A chatbot is a computer program created to simulate conversations with humans, typically through text or voice interactions. These programs are designed to assist users by providing quick and accurate responses, solving their questions, or guiding them through specific tasks or processes. Chatbots are widely used across many industries such as customer service, education, healthcare, and e-commerce because they offer great efficiency and can operate 24/7 without breaks.

There are mainly two types of chatbots. Rule-based chatbots operate by following pre-set rules and scripts, which allow them to handle simple and structured questions effectively. However, they struggle to manage more complicated or unpredictable conversations. On the other hand, AI-powered chatbots use artificial intelligence, including natural language processing (NLP) and machine learning, to provide smarter, more human-like responses. These AI-driven chatbots learn from user interactions and improve their performance over time. Hybrid chatbots combine the strengths of both rule-based and AI-powered systems, providing the reliability of structured rules while being flexible enough to

handle complex tasks. Additionally, voice-activated chatbots use speech recognition technology to process spoken commands, enabling hands-free interaction.

Chatbots have transformed the way people communicate digitally by making information more accessible, reducing waiting times, and efficiently handling repetitive tasks. Their flexibility and usefulness also make them highly valuable in educational settings, where they help students by answering questions, simplifying processes, and improving overall engagement.

Artificial Intelligence (AI) in Customer Service

The literature on Artificial Intelligence (AI) reveals its significant impact on customer service, providing scalable, real-time solutions that improve both customer experience and operational efficiency. AI systems, such as chatbots and virtual assistants, are designed to handle routine tasks, allowing human agents to focus on more complex issues (Khalil, 2023). Key advantages of AI include its ability to scale, enabling systems to handle multiple interactions at once while maintaining consistent service quality, even during peak times like holidays or product launches (Yang, 2023). Additionally, AI enhances decision-making through algorithms that analyze customer queries, route them effectively, and generate predictive insights to inform service strategies (Heinlein & Huchler, 2023). Natural Language Processing (NLP) and sentiment analysis further improve customer satisfaction by adapting responses to the tone and context of customer

messages. These techniques help AI provide personalized recommendations, analyze customer behavior, and predict future trends, positioning AI as a critical tool for modern customer service.

AI's ability to provide personalized recommendations, analyze customer data, and predict future trends underscores its pivotal role in modern customer service. For example, machine learning models can identify patterns in customer behavior, allowing AI systems to anticipate user needs and suggest products or services aligned with their preferences. This combination of scalability, real-time responses, and predictive insights ensures businesses can offer high-quality service while simultaneously reducing operational costs.

The literature underscores the transformative role of AI in customer service. Chatbots and virtual assistants powered by AI allow businesses to automate routine interactions, providing scalable and real-time responses to customer inquiries (Khalil, 2023). AI systems are particularly valuable in managing high volumes of interactions without compromising quality, ensuring customer satisfaction during high-demand periods (Yang, 2023). Furthermore, AI algorithms, including those used for query classification and predictive insights, contribute to the improvement of service strategies by analyzing customer interactions (Heinlein & Huchler, 2023). These developments have enhanced operational performance and allowed businesses to better meet modern customer expectations.

Empirical research highlights AI's effectiveness in customer service. Heinlein & Huchler (2023) demonstrated how AI-driven chatbots reduce wait times

and provide quick resolutions to customer queries, improving satisfaction. Khalil (2023) emphasized AI's role in query classification, which minimizes human error and ensures accurate, timely responses. Yang (2023) explored the scalability of AI systems, showing their ability to handle large volumes of interactions without sacrificing service quality during peak demand. Additionally, studies have highlighted how NLP and sentiment analysis enable AI to interpret customer emotions and respond in contextually relevant ways, leading to more personalized interactions (Heinlein & Huchler, 2023). These findings support AI's capacity to enhance customer service in both efficiency and quality.

AI is central to the development of ADAPT: Developing an AI-Driven Inquiry-Based Chatbot with Plugin Integration and Customization for Web In conclusion, AI is integral to the development of ADAPT: AI-Driven Inquiry-Based Chatbot with Plugin Integration and Customization for Web Systems. By leveraging supervised and unsupervised learning models, ADAPT can manage routine inquiries, provide real-time responses, and improve customer satisfaction while reducing the burden on human agents. The use of deep learning allows the chatbot to process unstructured data, identify patterns, and provide dynamic, context-aware responses. As ADAPT continuously learns from user behavior, it adapts to evolving needs, offering personalized solutions and enhancing customer interactions across diverse web platforms. The goal of this study is to utilize these AI capabilities to create a chatbot that not only improves user interactions but also

optimizes operational efficiency, delivering a more dynamic and satisfying customer experience.

Machine Learning (ML) in Chatbots

Machine Learning (ML) plays a pivotal role in enhancing the functionality of modern chatbots by enabling them to understand, respond to, and anticipate user queries more effectively. Supervised learning, one of the key ML techniques, involves training algorithms on labeled datasets where the correct responses are pre-identified. This technique is widely used for tasks like intent recognition and language understanding, allowing chatbots to process user queries accurately (Chen et al., 2023). Unsupervised learning, on the other hand, works with unlabeled data, identifying patterns and relationships within the dataset. It is particularly effective for clustering, anomaly detection, and adapting to dynamic conversations where predefined labels are unavailable, helping chatbots manage complex interactions (Luan & Tsai, 2021). Together, these ML techniques empower chatbots to improve their performance over time, making them adaptable and efficient for various applications.

The literature emphasizes the transformative role of machine learning in chatbot development, specifically through supervised and unsupervised learning techniques. Supervised learning, which involves training on labeled data, is crucial for intent recognition and language understanding in chatbots, enabling them to handle user queries accurately (Chen et al., 2023). In contrast, unsupervised

learning focuses on identifying hidden patterns in large datasets, providing chatbots with the flexibility to adapt to evolving user behaviors and complex conversations without the need for labeled data (Luan & Tsai, 2021). These learning techniques form the backbone of chatbot development, allowing AI systems to continuously improve and personalize interactions.

Empirical research has demonstrated the significant impact of ML on chatbot functionality. Adamopoulou & Moussiades (2020) illustrated how supervised learning enhances intent recognition by classifying user queries with high accuracy. Additionally, Chen et al. (2023) explored how supervised learning improves language understanding, enabling chatbots to provide contextually relevant responses. On the other hand, Luan & Tsai (2021) highlighted the role of unsupervised learning in identifying user preferences and adapting to changing needs, even in environments with limited labeled data. These studies underline the effectiveness of both supervised and unsupervised learning in enhancing chatbot performance and adaptability.

Machine Learning is a foundational component in the development of ADAPT: Developing an AI-Driven Inquiry-Based Chatbot with Plugin Integration and Customization for Web Systems. Supervised learning enables ADAPT to recognize user intents and deliver accurate responses based on historical data, ensuring reliability in interactions. Unsupervised learning, on the other hand, allows the chatbot to identify new patterns, handle diverse conversations, and adapt to dynamic environments. By leveraging these ML techniques, ADAPT aims to

enhance user interactions and provide personalized, contextually relevant responses in web systems. This combination of supervised and unsupervised learning makes ADAPT a versatile and effective tool in improving user satisfaction and engagement.

Deep Learning

Deep learning is a powerful subset of artificial intelligence that uses artificial neural networks to learn from data, as highlighted in the literature. These networks consist of layers of interconnected nodes, each designed to recognize specific features within the data. For example, in image recognition systems, the initial layers detect edges, subsequent layers identify shapes, and deeper layers recognize objects. During data processing, the network iteratively adjusts the connection weights between nodes, refining its accuracy in classification. This training process employs techniques such as supervised learning, unsupervised learning, or reinforcement learning, allowing the network to make predictions or decisions when presented with new data (Google Cloud, n.d.). Deep learning's capability to process information through artificial neural networks, mimicking the structure of the human brain, ensures it can analyze data in a non-linear fashion, much like how humans perceive and interpret the world (Staff, 2024).

Studies on deep learning emphasize its advantage in handling large, diverse, and unstructured datasets compared to traditional machine learning approaches. Unlike traditional methods, which often rely on human intervention to correct

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	errors, deep learning algorithms improve performance autonomously through repeated exposure to data. Research highlights that artificial neural networks excel in processing unstructured information, enabling applications like image recognition, language processing, and conversational AI to achieve significant advancements (Staff, 2024). These studies demonstrate how deep learning leverages iterative learning to recognize patterns and adapt dynamically, offering scalable solutions for complex data analysis tasks. Deep learning plays a pivotal role in ADAPT: AI-Driven Customizable Chatbot Plugin by enabling the chatbot to analyze and learn from extensive datasets, resulting in more personalized and contextually relevant user interactions. By integrating deep learning capabilities, ADAPT continuously refines its ability to process diverse user inputs and deliver accurate, real-time responses. Artificial neural networks empower ADAPT to recognize patterns and adapt to dynamic conversations, ensuring it can efficiently handle a wide range of user queries. As the chatbot learns from every interaction, it enhances its response quality over time, offering a more tailored and effective user experience. This adaptability is essential for web-based platforms, where user needs often evolve, and the ability to meet these changing demands ensures ADAPT remains a robust and reliable tool.	
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Chatbots

The development of AI-driven customizable chatbots has transformed user interaction on web-based platforms. These chatbots engage users in real-time, offering immediate responses to inquiries and improving information dissemination. Machine learning (ML) and Natural Language Processing (NLP) techniques are fundamental in enabling chatbots to understand and process user inputs. Supervised learning methods are commonly used to train chatbots on large datasets, enhancing their ability to classify intents and respond contextually (Bautista, 2020). Additionally, deep learning algorithms, such as Transformer models, are applied to improve response generation and sentiment analysis (Merritt, 2024). Support Vector Machines (SVM) and Logistic Regression are used for intent classification, while APIs allow chatbots to integrate external data, expanding their functionality and enhancing responsiveness (McIlravey, 2023). These advanced techniques and algorithms ensure that chatbots provide personalized, adaptive, and dynamic responses, significantly improving user experience.

In the literature, studies have highlighted the pivotal role of chatbots in enhancing user interaction and streamlining information dissemination in various domains. These AI-driven applications are designed to engage users in real-time, providing immediate responses to inquiries and facilitating seamless exchanges of information. The integration of chatbots into web development has been widely discussed, with an emphasis on their benefits, particularly in terms of customization and API integration, making them indispensable in a variety of industries.

Studies emphasize that customizable chatbots allow organizations to tailor interactions based on user preferences and behaviors, leading to improved engagement and satisfaction. Researchers underscore the importance of these features in enhancing user experiences. For example, chatbots equipped with Application Programming Interfaces (APIs) can access external data and services, enriching their responsiveness and functionality. Literature on the convergence of AI, bots, and metadata suggests the transformative potential of chatbots in creating a "whole new information world" (Irvin, 2023). Furthermore, APIs allow chatbots to address complex queries and provide comprehensive support, positioning them as essential tools in modern communication systems. The ability of chatbots to adapt communication styles further highlights their flexibility in addressing diverse user needs and ensuring meaningful interactions across different domains. These capabilities enable chatbots to serve users in ways beyond basic query resolution, incorporating personalization and dynamic responses that significantly enhance the user experience.

Empirical studies also illustrate the effectiveness of chatbots in specific contexts. For instance, Hsu et al. (2023) demonstrated that chatbots with expert decision-making mechanisms significantly enhance educational outcomes by personalizing responses and offering contextually relevant information. This approach has been shown to reduce user anxiety and improve achievement in educational settings. In addition, Olk, Tscheulin, and Zogaj (2020) examined the impact of communication styles used by COVID-19 chatbots in crisis

communication. Their findings reveal that tailored communication strategies improve public trust and facilitate effective information dissemination during emergencies. These studies highlight the value of chatbots in high-stakes scenarios, where customization and adaptability are crucial in meeting user expectations and achieving communication goals.

In conclusion, the literature and studies reviewed confirm the significant role of AI-driven customizable chatbots in improving user interaction across web-based platforms. By leveraging machine learning, NLP, and advanced algorithms like SVM and Transformer models, chatbots can deliver personalized, adaptive responses that enhance user engagement and satisfaction. The integration of APIs further enriches their functionality by enabling seamless interaction with external data. This research, ADAPT: AI-Driven Customizable Chatbot Plugin for Enhanced User Interaction in Web-Based Platforms, aims to explore these technologies in depth and develop a chatbot system that adapts to user needs, delivering efficient, personalized experiences across different domains.

Natural Language Processing (NLP)

Natural Language Processing (NLP) plays a crucial role in artificial intelligence by enabling machines to understand, interpret, and generate human language. NLP relies on techniques like tokenization, part-of-speech tagging, and named entity recognition to ensure smooth human-machine interaction (Montejo-Ráez & Jiménez-Zafra, 2022). Algorithms such as these are integrated into

frameworks like SpaCy, which simplifies NLP tasks such as language parsing and named entity recognition. Natural Language Generation (NLG) is another important aspect of NLP, where AI generates coherent text from structured data (Egger & Gokce, 2022). These techniques make NLP essential in various applications, including language translation, sentiment analysis, and content creation, helping AI systems understand context and respond appropriately to user inputs. In the context of the ADAPT AI-driven chatbot, these NLP tools enhance the system's ability to process user queries, adapt responses, and improve engagement on web-based platforms.

NLP's significance in enhancing communication between humans and machines is well-documented. Researchers emphasize how NLP techniques enable AI systems to better interpret and generate human language, thereby improving the quality and efficiency of communication across multiple domains (Egger & Gokce, 2022). NLP tools, such as word segmentation and dependency parsing, are critical in making interactions smoother and more meaningful, ensuring AI systems respond effectively to user needs (Montejo-Ráez & Jiménez-Zafra, 2022). These advancements allow chatbots and AI-driven applications to engage users more naturally, making them indispensable in digital platforms for enhancing user experience.

Empirical studies highlight the growing importance of NLP in AI applications. For instance, **Egger & Gokce (2022)** demonstrate how NLP is used to improve human-computer communication by enabling AI systems to analyze and

generate human language more efficiently. Likewise, **Montejo-Ráez & Jiménez-Zafra (2022)** showcase how SpaCy's pre-trained models facilitate complex language processing tasks, like named entity recognition, making it easier to integrate NLP into AI systems. These studies underline how NLP enhances the functionality and adaptability of AI systems, such as chatbots, across different platforms.

NLP is also a foundational component in the development of ADAPT, an AI-driven inquiry-based chatbot designed for web systems. By leveraging NLP tools such as word segmentation and named entity recognition, ADAPT can effectively interpret user queries and provide accurate, relevant responses. This functionality enhances the chatbot's interactivity and usability, making it an effective tool for improving user engagement and satisfaction. Furthermore, by incorporating insights from NLP research, ADAPT aims to not only facilitate efficient communication but also personalize interactions to meet the specific needs of academic platforms, thus ensuring a more tailored and effective user experience.

API Building

In the literature, APIs (Application Programming Interfaces) are defined as a set of protocols or rules that enable different software applications to communicate and share data, features, and functionalities. By streamlining the development process, APIs allow developers to integrate external capabilities rather than building them from scratch, saving time and resources. Furthermore,

APIs enhance security by sharing only necessary information, thus minimizing data exposure and protecting internal systems. They enable servers or devices to exchange relevant data packets instead of full datasets. Well-crafted API documentation plays a crucial role by serving as a technical guide that details an API's features and provides developers with clear instructions for interaction, ultimately contributing to the API's success (Goodwin, 2024).

Studies, such as those by Sonas Systems (n.d.), underscore the challenges and benefits of constructing APIs, describing the process as time-intensive but highly valuable for businesses of all sizes. These studies emphasize that APIs facilitate system integration, reducing human error through automation and enabling third-party developers to create custom solutions. Platforms offering APIs are increasingly popular due to their flexibility, allowing businesses to adapt their offerings to various needs and external systems.

The role of APIs in the development of ADAPT: AI-Driven Customizable Chatbot Plugin is indispensable, as they enable seamless communication between the chatbot and external systems, ensuring access to real-time data for contextualized and personalized responses. By integrating APIs, ADAPT can connect to various web-based platforms securely, maintaining privacy by exchanging only relevant information. This flexibility also allows the chatbot to scale efficiently by integrating with new services without requiring extensive redesign. Additionally, well-documented APIs simplify the development and

customization process, ensuring that ADAPT can adapt to diverse user needs and continuously evolve for more dynamic and efficient interactions.

Web-based UI and UX

User experience (UX) encompasses the entire interaction a person has with a product or service, from the initial engagement to the outcome. The primary goal of UX design is to create user-centered, intuitive, and visually appealing experiences that meet the needs and expectations of users. In the context of eCommerce, the "user" refers to customers interacting with a website or shopping app, and "experience" covers everything from the details, like the design of a shopping cart icon, to broader elements such as the ease of the checkout process. UX design focuses on optimizing every touchpoint of the user's journey, including digital platforms and even physical elements, like product packaging, to ensure a seamless and enjoyable interaction.

User interface (UI), a critical subset of UX, specifically deals with the visual and interactive components of a digital product or software application. UI is concerned with the design and functionality of the interface, ensuring that users can easily complete tasks and navigate the system. By applying principles such as branding, typography, color theory, and user research, UI enhances the overall user experience. Key UI elements, including buttons, icons, search bars, progress bars, animations, and feedback messages, all play a role in making the design user-friendly and engaging (Shopify Staff, 2023).

User Experience (UX) is fundamental to the development of ADAPT: AI-Driven Customizable Chatbot Plugin, as it ensures that users engage with the chatbot in a seamless, intuitive, and enjoyable manner. By optimizing the entire interaction, from the chatbot’s visual design to its responsiveness, ADAPT creates an efficient and user-friendly experience. User Interface (UI) contributes to this process by ensuring that the chatbot's interactive elements—such as buttons, icons, and feedback messages—are easy to use and navigate. Incorporating design principles like branding, typography, and color theory, ADAPT’s UI helps create an engaging environment where users can quickly access the information they need while enjoying a pleasant and intuitive interaction.

API Integration

Application Programming Interfaces (APIs) are crucial for connecting chatbot systems with web platforms, enabling seamless data exchange and functionality (Lee et al., 2024). By facilitating communication between disparate software applications, APIs empower chatbots to access external resources, enhance user interactions, and streamline operations (Cruz et al., 2014). Understanding how APIs function is essential for leveraging their potential in developing robust and versatile chatbot solutions.

APIs operate through a set of protocols and tools that define how software components should interact (Holtmanns, 2004). In chatbot systems, APIs enable various functionalities, such as retrieving user information, processing requests,

and integrating with third-party services (Lee et al., 2024). This functionality significantly enhances the chatbot's versatility, allowing it to handle complex tasks and provide real-time data access (Holtmanns, 2004). For example, integrating with customer relationship management (CRM) systems allows chatbots to retrieve customer data and offer personalized responses, which improves user experience and satisfaction (Lee et al., 2024).

Moreover, APIs enable chatbots to support multiple communication channels, including websites, mobile apps, and messaging platforms like WhatsApp and Facebook Messenger (Cruz et al., 2014). This omnichannel approach ensures that users can engage with the chatbot through their preferred platforms, creating a cohesive and accessible experience (Lee et al., 2024). As organizations strive to meet users where they are, APIs facilitate flexibility, allowing chatbots to maintain consistent dialogue across various interfaces (Holtmanns, 2004).

However, while API integration offers numerous benefits, it also introduces security risks. The open nature of APIs can expose systems to vulnerabilities if proper authentication and security measures are not in place (Holtmanns, 2004). To mitigate these risks, robust authentication mechanisms, such as OAuth tokens, API keys, and secure login processes, are essential (Cruz et al., 2014). These measures ensure that only authorized users can access sensitive information, creating a secure environment for interactions and fostering trust in the chatbot system (Lee et al., 2024).

Implementing a secure login/registration module is a key aspect of securing API interactions in chatbot systems (Cruz et al., 2014). By requiring users to authenticate before accessing specific features, organizations can protect user data and maintain control over interactions (Holtmanns, 2004). A secure login process not only safeguards information but also builds user trust, encouraging more confident and frequent engagement (Lee et al., 2024).

In addition to security, the performance of API integrations is critical. Latency issues can arise if APIs are not optimized, leading to slow response times that frustrate users (Holtmanns, 2004). To address this, developers must ensure that API calls are efficient and that backend services can handle the expected load (Lee et al., 2024). Implementing caching for frequently requested data can also improve performance by reducing the need for repeated API calls (Cruz et al., 2014).

In conclusion, API integration is a foundational element of modern chatbot systems, enabling enhanced functionality and user engagement (Lee et al., 2024). APIs allow chatbots to access external data and services, delivering personalized and contextually relevant interactions (Holtmanns, 2004). However, ensuring the security and performance of these integrations is paramount. By employing robust authentication mechanisms, protecting against vulnerabilities, and optimizing performance, developers can create secure and effective chatbot solutions that enhance user experience (Cruz et al., 2014). As API technology evolves, it promises further enhancements, making chatbots increasingly integral to digital communication strategies (Lee et al., 2024).

API integration is a key aspect of ADAPT: AI-Driven Customizable Chatbot Plugin, allowing the chatbot to connect with various web platforms and external services. By leveraging APIs, ADAPT can access real-time data, offer personalized responses, and integrate with third-party systems such as CRM tools. This enhances both the functionality and the user experience, making the chatbot more versatile and efficient for web-based systems. Ensuring secure and optimized API integration will be crucial for delivering a reliable and responsive chatbot.

The API integration in ADAPT ensures seamless interaction with web platforms and third-party services, enabling real-time data retrieval and personalized responses. APIs enhance the chatbot's ability to handle complex tasks and engage users across multiple channels, including websites, mobile apps, and messaging platforms. However, securing these integrations is essential. Proper authentication mechanisms, like OAuth tokens and secure login processes, protect user data and maintain trust in the system. Performance optimization, including reducing latency and implementing caching, ensures that ADAPT remains responsive and efficient. API integration allows ADAPT to provide a dynamic, secure, and personalized user experience, making it a highly effective solution for web-based platforms.

System Testing and Performance Evaluation

In the software development lifecycle, system testing, and performance evaluation are essential stages that guarantee a system performs as planned under a

variety of circumstances. System testing entails confirming that a software application's integrated components work as intended. Functional testing, which verifies certain features, and non-functional testing, which evaluates factors like security, scalability, and usability, are included in this step. One important part of non-functional testing is performance evaluation, which measures the system's performance under load, stress, and different operating situations. Metrics including response time, throughput, and resource usage are evaluated as part of this. A well-executed performance evaluation facilitates the identification of bottlenecks, guarantees system stability, and confirms that the system is capable of handling expected user traffic and processing demands. To provide dependable, effective, and scalable software and assist companies in meeting quality requirements and user expectations prior to deployment, system testing, and performance evaluation are essential.

SCRUM Methodology in Software Development

SCRUM is an agile approach that is used in software development for managing intricate projects with adaptability and incremental development. It places a strong emphasis on a team-based methodology in which teams develop, test, and deploy software in "sprints"—time-boxed iterations that usually span two to four weeks. Every sprint allows for ongoing input and adjustments since it starts with planning and concludes with a review.

The requirements and priorities can change in response to feedback from stakeholders during the process, the system promotes adaptability. The Product Owner, who sets priorities for the product backlog, the SCRUM Master, who makes sure the team adheres to SCRUM procedures, and the development team, which oversees meeting sprint targets, are important positions in SCRUM.

The SCRUM methodology is highly relevant to our thesis, ADAPT: Developing an AI-Driven Inquiry-Based Chatbot with Plugin Integration and Customization for Web Systems, as it allows us to manage the development process in an adaptive and incremental way. By using SCRUM, our team can break down the complex tasks involved in building and integrating the chatbot into shorter sprints, allowing for continuous feedback and adjustments. This iterative approach ensures that we can refine features like API integration, NLP capabilities, and user interaction in response to stakeholder input, ultimately delivering a more efficient and user-centered chatbot system.

SCRUM improves productivity, guarantees frequent feedback loops, and boosts customer satisfaction by segmenting huge projects into smaller, more manageable units. This allows features to be delivered incrementally rather than waiting for the whole product launch (Schwaber & Sutherland, 2020). Teams can address risks early and continuously improve procedures and deliverables thanks to their iterative nature.

Back-End Technologies

Back-end technology is a foundational component of web and software applications, powering essential processes such as server-side logic, database interactions, and front-end communication. Key programming languages, including Java, Python, Ruby, and Node.js, are commonly used to build the server logic that responds to and processes user requests. Databases like PostgreSQL, MongoDB, and MySQL are crucial for storing and managing data, ensuring that information can be retrieved and updated swiftly. Additionally, server frameworks such as Django, Express.js, and Spring simplify back-end development by providing reusable components for routing, session management, and security.

Incorporating web services and APIs into the back end allows communication between the server and external systems or services, further enhancing the application's functionality. By managing the behind-the-scenes operations, back-end development ensures that applications are scalable, secure, and efficient, providing a seamless user experience (Lawrence & Abraham, 2021). This layer is essential for ensuring that apps perform reliably, even under high demand, and can operate across various platforms.

For ADAPT: AI-Driven Customizable Chatbot Plugin, back-end development plays a vital role in managing server-side logic, handling database interactions, and facilitating communication between the front-end and external systems. By utilizing programming languages like Python and Node.js, ADAPT can process user requests in real-time and efficiently manage interactions. The

integration of databases such as PostgreSQL or MySQL enables rapid storage and retrieval of user data, allowing the chatbot to deliver dynamic and personalized responses. Furthermore, back-end web services and APIs allow ADAPT to integrate seamlessly with third-party platforms, such as customer relationship management (CRM) systems, enhancing the chatbot's functionality.

The adoption of back-end frameworks like Django or Express.js ensures that ADAPT is scalable, secure, and capable of handling high traffic volumes, thus delivering a reliable and smooth user experience. Robust back-end development guarantees that ADAPT operates efficiently, processes requests accurately, manages data securely, and maintains functionality across various platforms.

Front End Technologies

Front-end technologies are crucial for the development of dynamic, user-friendly web applications. According to Freeman & Robson (2022), these technologies are essential for creating responsive, interactive, and accessible applications, as they focus on the layout, interaction, and design that users engage with directly. In the context of ADAPT: AI-Driven Customizable Chatbot Plugin, HTML (Hypertext Markup Language), CSS (Cascading Style Sheets), and JavaScript play a fundamental role in ensuring a seamless user experience. HTML helps define the structure of the web pages, providing the foundation upon which the layout and content are built. CSS ensures a visually appealing and responsive

design, and JavaScript enables dynamic behavior, allowing ADAPT to interact with users in real-time, as outlined by Freeman & Robson (2022).

Additionally, modern JavaScript frameworks and libraries, such as React, Angular, and Vue.js, offer developers the ability to create dynamic user interfaces by providing reusable components and efficient state management. This is important for ADAPT, as React, for example, allows for real-time updates, enabling the chatbot to deliver personalized responses efficiently. This flexibility is emphasized in a study by Freeman & Robson (2022), who highlight that JavaScript frameworks streamline the development of complex applications while ensuring a smooth user experience. By leveraging these technologies, ADAPT can create a highly responsive interface that reacts to user input in real-time.

Moreover, CSS frameworks like Tailwind CSS and Bootstrap are crucial for ensuring that ADAPT is visually consistent and functional across a range of devices. As noted by Freeman & Robson (2022), CSS frameworks simplify the design process, guaranteeing that web applications look and perform well on various screens. ADAPT can easily adapt to different device sizes, maintaining both design aesthetics and usability. These frameworks help ensure that the chatbot plugin's interface is responsive and accessible, which is vital for delivering an optimal user experience across platforms.

To further optimize development workflows, front-end build tools like Webpack, Babel, and npm are indispensable. These tools manage project dependencies, ensure compatibility across browsers, and improve code

performance. According to Freeman & Robson (2022), build tools like Webpack are crucial for bundling and optimizing code, ensuring faster load times, while Babel provides compatibility for different browsers by transpiling modern JavaScript. npm helps manage these dependencies and packages efficiently, streamlining the overall development process for ADAPT and improving its performance.

In summary, integrating these front-end technologies and frameworks ensures that ADAPT is both functional and engaging. By applying technologies such as HTML, CSS, JavaScript, React, Tailwind CSS, and Webpack, ADAPT can deliver a seamless, dynamic, and responsive user experience. These technologies work together to create an interactive, accessible, and user-friendly interface that enhances the chatbot's capabilities and ensures it meets the needs of users across multiple platforms, as highlighted by Freeman & Robson (2022).

AI in E-Commerce Websites in the Philippines A study by Mercado (2021) focused on the integration of AI-driven chatbots on e-commerce websites in the Philippines. It showed that small and medium enterprises (SMEs) have seen significant improvements in customer engagement due to these AI tools. The use of chatbots provided 24/7 support, reducing human labor costs while improving customer satisfaction (Mercado, 2021).

AI and User Experience in Philippine Government Websites In a 2022 study by Dela Cruz, the role of AI in enhancing user experience on Philippine government

websites was explored. The research highlighted how AI-driven data processing improved accessibility and navigability, ensuring that government services were more efficiently delivered online, especially in response to the COVID-19 pandemic (Dela Cruz, 2022).

The Role of Machine Learning in Filipino Online Education Platforms
Bautista's (2020) research examined the use of machine learning algorithms in Filipino online education platforms. It emphasized how personalized learning and adaptive testing, powered by AI, significantly enhanced student engagement and learning outcomes in various web-based learning systems (Bautista, 2020).

AI and Online Consumer Behavior in the Philippine Retail Industry Reyes (2019) explored how AI-based recommendation systems on retail websites influenced consumer behavior in the Philippines. The study revealed that personalized product suggestions driven by machine learning algorithms resulted in higher conversion rates for online businesses, improving both customer satisfaction and sales (Reyes, 2019).

AI-Driven Analytics in Philippine News Websites Garcia's (2021) study focused on the integration of AI-based analytics in Filipino news websites. The research underscored how AI algorithms were used to track user behavior, providing news organizations with insights into audience preferences and helping them deliver more relevant content through personalization techniques (Garcia, 2021).

Natural Language Processing in Filipino Social Media Websites Mendoza (2020) investigated the application of Natural Language Processing (NLP) in moderating and managing user-generated content on Filipino social media platforms. The study highlighted how AI-powered NLP systems were employed to identify and filter inappropriate content, enhancing community management and online safety (Mendoza, 2020).

AI-Powered Digital Marketing in Philippine Startups Santos (2021) explored the growing adoption of AI in digital marketing by Philippine startups. The study found that AI-based tools, such as predictive analytics and automated content generation, helped these startups improve their online presence and engage better with potential customers, resulting in increased conversions and cost-efficiency (Santos, 2021).

AI in Philippine Healthcare Websites In a study by Cruz and Rivera (2022), the researchers examined how AI tools, such as symptom checkers and AI-driven virtual consultations were integrated into healthcare websites in the Philippines. The study highlighted how these tools improved accessibility to medical information and reduced the burden on healthcare professionals (Cruz & Rivera, 2022).

AI-Driven Recruitment Portals in the Philippines A study by Villanueva (2020) analyzed the role of AI in recruitment websites in the Philippines, focusing on how AI-driven algorithms streamlined the hiring process. The research emphasized that AI-powered resume screening and candidate matching reduced

recruitment time for companies, leading to more efficient hiring (Villanueva, 2020).

AI in Customer Service for Philippine Telecom Websites Perez (2019) researched the use of AI chatbots in Philippine telecom company websites. The study found that AI systems helped resolve customer queries more quickly and effectively, which improved user satisfaction and reduced operational costs for telecom firms (Perez, 2019).

AI-Powered Web Design Automation in the U.S. A study by Smith et al. (2019) in the United States analyzed how AI-powered automation tools are revolutionizing web design. By enabling automated layout generation and content structuring, AI has streamlined the web development process, reducing the workload for designers while ensuring responsive and visually appealing websites (Smith et al., 2019).

The Impact of AI on Website Personalization in Europe A 2020 study conducted by Muller and Krause in Germany explored the impact of AI on website personalization. It demonstrated that AI algorithms could analyze user behavior and preferences, dynamically adjusting website content in real-time to enhance user engagement, improve user satisfaction, and increase site retention rates (Muller & Krause, 2020).

AI-Driven Recommendation Systems in Chinese E-Commerce Zhang (2021) studied the role of AI in recommendation systems on Chinese e-commerce platforms such as Alibaba. The research found that AI-driven algorithms, utilizing

data from user profiles and previous purchases, were crucial in making accurate product recommendations, which led to increased customer retention and sales growth (Zhang, 2021).

AI Chatbots and User Experience in UK-Based Financial Services A study by Johnson et al. (2020) analyzed the use of AI-driven chatbots in the financial sector of the UK. It revealed that AI chatbots, integrated into financial websites, improved user experience by providing instant support, answering queries, and handling routine banking transactions, thus saving time and increasing customer satisfaction (Johnson et al., 2020).

Machine Learning for Content Optimization in Australia According to Thompson (2022), a study in Australia focused on machine learning models used for content optimization in media and news websites. The findings highlighted that AI-driven content management systems helped predict user preferences, allowing website managers to prioritize trending topics and boost audience engagement (Thompson, 2022).

AI and Web Accessibility in Canadian Public Websites A study by Anderson and Lee (2021) in Canada examined how AI tools were used to improve web accessibility for people with disabilities. AI-driven technologies, such as voice recognition and automated text generation, made public websites more accessible to individuals with visual and auditory impairments, promoting inclusivity in online services (Anderson & Lee, 2021).

AI-Driven Content Moderation in Social Media Platforms (USA) Roberts (2020) examined the use of AI-driven content moderation tools on major social media platforms in the United States. The study highlighted how AI systems played a critical role in identifying and removing harmful content, ensuring a safer online environment for users (Roberts, 2020).

AI in Travel and Tourism Websites (France) Dubois (2021) explored how AI enhanced customer interactions on travel and tourism websites in France. The study highlighted how AI tools like virtual tour guides, personalized trip recommendations, and dynamic pricing models transformed user experiences, driving more personalized travel planning and bookings (Dubois, 2021).

AI in Online Customer Support in Japanese Retail Websites Saito (2019) analyzed the role of AI chatbots and virtual assistants in providing customer support on Japanese retail websites. The research showed that these AI tools significantly reduced customer wait times, improved customer satisfaction, and minimized the need for human intervention in routine inquiries (Saito, 2019).

AI for Web Security in the Banking Sector (India) A study by Patel and Kumar (2022) focused on the use of AI in enhancing web security for banking websites in India. The research showed that AI algorithms detected and prevented cyberattacks, such as phishing and fraud, improving the overall security and trustworthiness of online banking platforms (Patel & Kumar, 2022).

Synthesis

The integration of Artificial Intelligence (AI) in websites has rapidly transformed digital landscapes across various sectors, both locally in the Philippines and globally. Local studies, such as those by Mercado (2021) and Reyes (2019), demonstrate how AI-driven recommendation systems and chatbots significantly boost customer engagement and online sales for SMEs in the Philippines. In the education sector, Bautista (2020) shows that AI-powered adaptive learning systems improve students' engagement and learning outcomes on Filipino e-learning platforms. Similarly, AI-driven analytics have empowered Filipino news websites to deliver personalized content (Garcia, 2021), while AI chatbots have streamlined customer support on Philippine telecom websites (Perez, 2019). These examples reflect the powerful role AI plays in optimizing various aspects of website functionality, enhancing user experience and operational efficiency.

On the international front, studies show the wide-ranging applications of AI in web design, e-commerce, customer service, tourism, and web security. For instance, Smith et al. (2019) and Dubois (2021) demonstrate how AI automates web design and enhances user experience on travel and tourism websites, respectively. AI recommendation systems, particularly in large e-commerce platforms like Alibaba, have played a vital role in driving sales and improving user satisfaction, as seen in Zhang's (2021) research in China. Meanwhile, in the field of web security, Patel and Kumar (2022) highlight the effectiveness of AI in safeguarding

online banking systems from cyber threats in India, underscoring the critical role of AI in enhancing web-based services' trustworthiness. These global examples further underscore the adaptability and significance of AI across various digital sectors.

Both local and foreign studies consistently emphasize the importance of AI in optimizing user experiences, increasing engagement, and improving efficiency. For instance, AI-powered chatbots across sectors—from retail in Japan (Saito, 2019) to telecom websites in the Philippines (Perez, 2019)—help resolve customer inquiries faster and more effectively. Additionally, AI's role in content personalization is evident across sectors, as shown in Reyes (2019) for Philippine retail and Muller & Krause (2020) for European websites. Furthermore, studies on AI in content moderation (Roberts, 2020) and web security (Patel & Kumar, 2022) reveal the growing role of AI in ensuring safe and secure online environments. These studies highlight how AI not only enhances user interaction but also ensures the reliability and safety of digital services.

Overall, the literature reveals that AI's integration into websites brings forth a combination of efficiency, personalization, and improved user experience. Both local and global contexts highlight how AI's automation capabilities not only reduce operational costs but also enhance digital services across sectors like retail, education, finance, healthcare, and tourism. As AI continues to evolve, it is likely to become an even more integral part of web development, transforming the ways users interact with digital platforms. The widespread adoption of AI tools like

chatbots, recommendation systems, and personalized content demonstrates the tangible benefits AI offers in terms of increasing customer satisfaction and driving business success.

The studies show that AI is being widely used in different industries, both in the Philippines and around the world. AI tools like chatbots, recommendation systems, and personalized content are improving user experiences, increasing engagement, and making processes more efficient. For example, in the Philippines, AI has helped small businesses increase sales and improve customer service (Mercado, 2021; Reyes, 2019), and it has also improved student engagement in online education platforms (Bautista, 2020). Globally, AI is playing a key role in areas like e-commerce, cybersecurity, and tourism. In China, AI recommendation systems are driving customer loyalty and sales (Zhang, 2021), while in India, AI is improving the security of banking websites (Patel & Kumar, 2022). These examples highlight how AI is making things easier for both businesses and users by automating tasks, reducing costs, and improving customer satisfaction.

Surveys show that artificial intelligence is widely applied across many industries, both locally and internationally. AI tools like chatbots, recommendation engines, and personalized content are improving user experiences, boosting engagement, and making processes more efficient. For example, AI has improved customer service and increased sales for small businesses in the Philippines (Mercado, 2021; Reyes, 2019) and has enhanced student participation on online education platforms (Bautista, 2020). Globally, AI is playing a key role in industries

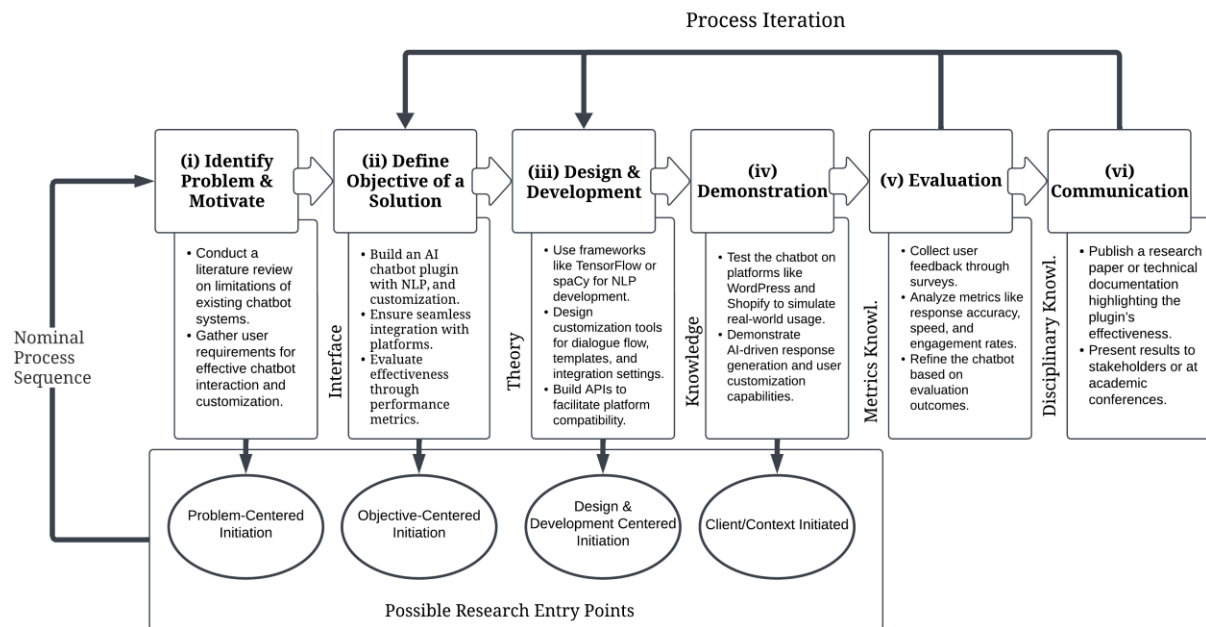
like cybersecurity, e-commerce, and tourism. In China, AI recommendation systems increase sales and customer loyalty (Zhang, 2021), while in India, AI strengthens the security of banking websites (Patel & Kumar, 2022). These examples highlight how AI helps lower costs, streamline tasks, and improve customer satisfaction.

However, there are some gaps in the research. While AI chatbots are used in many industries, there is little focus on their application in academic websites, especially chatbots that can be customized to fit specific needs. Most research explores general AI use in education, but it does not investigate how tailored AI chatbots could improve communication between students and academic systems. There is also limited research on how customizable chatbots could adapt to different university platforms. This study could explore how a personalized chatbot, designed to meet the specific needs of various institutions, could improve user experiences. This gap presents an opportunity for further investigation into how AI can be utilized more effectively in the academic sector.

Another challenge is the high cost of chatbot platforms, which can be expensive to implement. Many existing platforms also lack customizable widget options, limiting flexibility for users. Additionally, the hybrid approach of combining rule-based systems with machine learning for chatbot dialogue responses is not commonly addressed in research. This approach could lead to more accurate and efficient chatbot interactions. Addressing these gaps could help make chatbots more cost-effective and adaptable for academic settings, allowing

institutions to leverage AI more effectively while ensuring a positive user experience.

Conceptual Framework



Discussion of Design Science Research (DSR) Framework

In Phase 1 Identify Problem & Motivate: This step involves conducting a literature review to identify limitations in existing chatbot systems, such as poor customization and limited platform integration. User feedback is gathered to determine the necessary features and improvements, ensuring the chatbot plugin addresses these gaps.

In Phase 2 Define Objective of a Solution: This step, the chatbot is designed using advanced NLP techniques to ensure accurate, dynamic interactions.

Customization options like dialogue flows and templates are incorporated, and the plugin is tested for platform compatibility, focusing on performance metrics such as response accuracy and user engagement.

In Phase 3 Design & Development: In this phase, NLP tools like TensorFlow or spaCy are used to develop the chatbot's capabilities. Customization tools for dialogue flow and templates are designed, and APIs are created to ensure seamless integration across platforms, making the chatbot highly adaptable.

In Phase 4 Demonstration: The chatbot is tested on platforms like WordPress and Shopify to showcase its AI-driven response capabilities and customization features. This phase validates its real-world functionality, ensuring it meets user needs and integrates seamlessly across different environments.

In Phase 5 Evaluation: User feedback is collected through surveys, and key metrics like accuracy, speed, and engagement are analyzed to evaluate the chatbot's effectiveness. Based on this feedback, the chatbot is refined to enhance its functionality and user experience.

In Phase 6 Communication: The findings are compiled into a research paper or technical documentation, highlighting the chatbot's effectiveness and performance. Results are then presented to stakeholders or at conferences to communicate the success and impact of the project.

Definition of Terms

The definition of terms provides a deeper understanding of the key concepts, systems, and technologies used in this study. The following definitions will outline and explain these elements:

Natural Language Processing (NLP). It is a branch of artificial intelligence (AI) that uses machine learning that allows computers to read and write texts comparable to a human language.

Machine learning. It is a field of study in artificial intelligence (AI) that gives computers the ability to learn and improve on their own using data and without someone having to program it.

Natural Language Understanding (NLU). It is the computers' ability to understand or interpret human language. It analyzes the meaning of texts.

Chatbot. It is an automated computer program that interacts with customers engaging in conversation through a text messaging interface.

Application Programming Interface (API). It is a way for two software to communicate with each other. It transmits data from one program to another program.

Artificial Intelligence (AI). It refers to the field of computer science focused on creating systems and technologies that can perform tasks requiring human-like intelligence.

Intent. Within Natural Language Processing (NLP) and machine learning, an intent refers to the purpose or goal behind a user's input or request and

understanding what the user aims to achieve through their interaction with a system, such as a chatbot.

Entity. In the context of Natural Language Processing (NLP) and information retrieval, an entity refers to a unique piece of information or specific object identified within a text or data set.

Dialogue flow. Refers to the structured sequence of interactions between a user and a system. It outlines how the conversation progresses, including how the system responds to the user inputs and guides the interaction towards achieving a specific outcome.

System performance analysis. Involves evaluating the amount of work accomplished by a computer system.

CHAPTER III

METHODOLOGY

This chapter outlines the research methodology employed in the development and evaluation of "ADAPT: AI-Driven Customizable Chatbot Plugin for Enhanced User Interaction in Web-Based Platforms." The methodology is designed to ensure a systematic approach to gathering and analyzing data that will help determine the effectiveness of the ADAPT chatbot plugin in enhancing user interaction.

Research Design

This study will use a **quantitative research design** to evaluate the design, development, and performance of an AI-driven customizable chatbot plugin. The research will focus on measurable factors such as user feedback, engagement metrics, and system performance. The first phase, **Conceptualization and Literature Review**, will involve reviewing existing studies on chatbot technologies, NLP, AI, and system integration to identify gaps and create a framework for the plugin. The next phase, **System Development**, will include designing the architecture, developing the system using tools like Python and TensorFlow, and conducting functionality, performance, and security testing.

In the **Implementation Phase**, the chatbot will be integrated into platforms like WordPress and Shopify. Structured surveys with Likert scale questions will collect user feedback on satisfaction, usability, and system performance. Metrics such as response accuracy, engagement

rates, and response times will also be analyzed to assess the chatbot's operational efficiency. These structured methods will provide reliable, measurable data for evaluation.

The collected data will be analyzed using statistical software like SPSS or R. **Descriptive statistics** will summarize key metrics, while **inferential statistics** will identify significant differences in satisfaction and performance. This analysis will provide clear, data-driven insights into the chatbot's effectiveness and areas for improvement, helping guide future development and optimization.

Algorithms

This study will experiment with **Logistic Regression** and **Support Vector Machine (SVM)** algorithms for intent classification in the customizable chatbot system. Logistic Regression is a statistical method used for predicting categorical outcomes and is effective in multiclass classification tasks. It provides interpretable results by showing the relationship between input features and predicted intents. SVM, on the other hand, is a machine learning algorithm that separates classes using an optimal hyperplane and performs well with non-linear boundaries and high-dimensional data. A comparison of these algorithms will determine which provides better accuracy for classifying user intentions and improving chatbot interactions.

The **experimental analysis** will evaluate both algorithms using a labeled dataset that includes diverse user inputs to simulate real-life chatbot scenarios. The dataset will be split into training and testing subsets, with the training data used to build the models and the testing data for evaluation. Preprocessing steps, such as tokenization, vectorization, and normalization, will

prepare the data for algorithm training. Performance metrics like **accuracy**, **precision**, **recall**, and **F1-score** will be used to assess the models, with cross-validation ensuring robustness and avoiding overfitting.

A **comparative analysis** of the results will identify the strengths of each algorithm. The study hypothesizes that both Logistic Regression and SVM will perform well, but SVM may achieve higher accuracy due to its ability to handle complex decision boundaries and high-dimensional data. The findings will guide the choice of the most effective algorithm for improving chatbot user interactions.

Metrics

The authors will use the following number of key metrics to test each algorithm's performance in comparison:

Accuracy: This is the measure of how good the model is, taken by dividing the classified instances by the total. It just provides an overall general measure of how good the model performs.

Precision: It is the ratio of true positive predictions to all those that were predicted to be positive. In other words, it gives a ratio of how many of the predicted intents are relevant.

Recall: It is the ratio of true positive predictions to the total actual positives in the data set, which indicates just how good a model is at identifying all the instances in the data that are relevant for prediction.

F1-Score: Harmonic averages of precision and recall whose value will be quite preferable with a balance of the two metrics in cases of class imbalances.

Key Variables

Such analysis would include features abstracted from user inputs in the form of text vectors that may be generated through methods such as TF-IDF or word embeddings, along with hyperparameters chosen for each algorithm. Analyzing such metrics and variables systematically would give conclusive insights into whether Logistic Regression and SVM are effective for intent classification and guide us for further improvements in the capabilities of chatbot interactions.

Comparative Analysis

Comparative Analysis will involve a detailed comparison of the performance results achieved by both Logistic Regression and Support Vector Machine (SVM) algorithms to classify user intents. To begin with, the researchers will train and test both models on the same dataset. A comparison of the performance metrics, such as accuracy, precision, recall, and F1-score, will be established for each algorithm to find out which gives better classification results.

The researchers will present our findings in a clear, easy-to-understand manner and illustrate differences in performance using charts and graphs. We also investigate possible factors that could influence the effectiveness of each algorithm, such as the nature of input features or the complexity of user intents.

This analysis will not only state the strengths and weaknesses of each algorithm but at the same time contemplate how the best approach will be implemented within the customizable system of the chatbot.

To understand how these different models perform well in reality through this analysis can make it possible to provide recommendations so that the chatbot effectively meets user needs in the most efficient manner with excellent accuracy in classification. By the end, this comparative study will contribute vital insights for further improvement into intent classification methodologies in chatbot technology.

Sampling Technique

This study will use a **purposive sampling method** to select participants who are relevant to the research goals. The sample will include end-users, developers, and stakeholders who have interacted with chatbot plugins on platforms like WordPress, Shopify, or other e-commerce websites within the past six months. This ensures the feedback reflects recent user experiences.

Participants will be grouped based on their experience with chatbot technology: beginners, intermediate users, and advanced users. Developers will be included to provide technical insights, while stakeholders will assess how well the system meets business needs. The study will aim to include at least **50 participants** for surveys. Recruitment will take place through online forums, professional networks, and direct outreach to companies using chatbot plugins.

This approach ensures the data collected is relevant and statistically significant. By focusing on participants with direct experience, the study will provide a broad and accurate

understanding of chatbot usage and performance. The findings will help evaluate the chatbot's effectiveness and identify areas for improvement based on measurable feedback.

Participants of the study

The table below outlines the different groups of participants involved in this study, categorized based on their roles and selection criteria. The first group, **Customers/Chatbot Users**, includes 80 individuals who have interacted with chatbot plugins on platforms like e-commerce and customer service websites within the last six months. The second group, **Developers**, consists of 10 software professionals with experience in building, integrating, or maintaining chatbot plugins. Lastly, the **Business Stakeholders** group comprises 10 decision-makers from organizations that use chatbot plugins to improve customer engagement or streamline operations.

These groups were selected to ensure a well-rounded perspective on chatbot usage and effectiveness. The **Customers/Chatbot Users** provide insights into usability and user experience, while the **Developers** offer technical expertise on system functionality. The **Business Stakeholders** contribute a strategic viewpoint, assessing the impact of chatbot plugins on organizational goals. This diverse participant base ensures the study captures technical, functional, and experiential aspects of chatbot integration.

Participant Group	Number of Participants	Role	Selection Criteria
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(Customers /Chabot Users)	80	Users interacting with chatbot plugins	Individuals who have used chatbot plugins on web-based platforms (e.g., e-commerce, customer service sites) within the past six months.
Developers	10	Software Developers	Professionals with experience in designing, integrating, or maintaining chatbot plugins.
Business Stakeholders	10	Decision-makers utilizing chatbots	Stakeholders from organizations employing chatbot plugins for customer engagement or operational support.

End-Users. End-users, as the primary consumers of chatbot functionalities, form the largest participant group. They interact directly with chatbot systems, providing valuable insights into their usability, user satisfaction, and overall experience. The inclusion of individuals from varied industries such as e-commerce, customer support, and information services ensure a comprehensive evaluation of different use cases. Their feedback is instrumental in identifying usability challenges and opportunities for improvement.

Developers. Developers contribute crucial technical perspectives on chatbot plugin integration. This group comprises professionals with significant experience in designing

and maintaining chatbot systems for web-based environments. Their expertise provides insights into challenges related to system compatibility, performance optimization, and technological implementation. This technical input is vital for addressing operational inefficiencies and advancing chatbot design.

Business Stakeholders. Business stakeholders offer a managerial viewpoint, focusing on the strategic application and impact of chatbot plugins. This group includes decision-makers responsible for assessing the value of chatbots in enhancing customer engagement, improving operational efficiency, and achieving organizational goals. Their feedback highlights the broader implications of chatbot adoption, including cost-effectiveness, user engagement metrics, and organizational benefits.

The combination of these participant groups ensures that the study addresses all critical dimensions of chatbot integration, from user experience and technical functionality to strategic business considerations. This comprehensive approach facilitates actionable insights and contributes to the development of more effective and user-centric chatbot systems.

The study will involve a chatbot manager and its corresponding end users. The chatbot manager will be responsible for managing the chatbot system and will share feedback about how it works and how easy it is to manage. The end users will come from different web-based platforms that use chatbots, such as online shopping sites, customer service pages, and educational platforms.

The participants will be between 18 and 50 years old and will have different levels of technical skill to ensure a wide range of potential chatbot users. Each end user will spend time

using the ADAPT chatbot. Their feedback will be recorded, evaluated and analyzed to measure things like engagement and satisfaction.

By including a diverse group of participants and using a mixed-method approach, the study aims to see if the ADAPT chatbot can improve user interaction on web-based platforms.

Research Locale

his study will be conducted at the **Lyceum of the Philippines University (LPU) – Cavite Campus** in General Trias, Cavite, Philippines. The campus provides a diverse environment of students, faculty, staff, and administrators, making it an ideal setting for testing and evaluating the customizable chatbot system. The study will focus on two departments: the **Communication and Public Affairs Department (CPD)** and the **Center for Information Technology and Digital Learning (ITD)**. These departments were selected due to their relevance to chatbot usage. The CPD handles marketing and stakeholder inquiries, offering a real-world environment to test the chatbot's ability to manage communication tasks. Meanwhile, the ITD manages digital systems and infrastructure, providing the technical support needed for effective testing.

Testing will take place in the CPD and ITD offices, where participants can engage with the chatbot under practical conditions. These environments are equipped with reliable internet access and necessary technological infrastructure. The participants will include end-users, developers, and business stakeholders, who will provide insights into the chatbot's usability, functionality, and strategic benefits. Coordination with department heads and established communication channels will facilitate participant access and ensure an efficient recruitment process.

	LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE	
	By conducting the study at LPU Cavite, the research ensures that both the technical performance and user experience of the chatbot are evaluated in a dynamic academic setting. The diverse perspectives from participants will help capture a comprehensive understanding of chatbot integration, aiding in the development of a system that meets practical needs effectively. Research Instrument The research instrument for this study will collect quantitative data to assess the chatbot's effectiveness in improving user interaction, engagement, and satisfaction. Structured questionnaires will be the primary tool, featuring Likert scale questions to measure satisfaction, usability, and effectiveness, along with rating scale questions to evaluate performance and feature usefulness. These questionnaires are designed to generate measurable responses for statistical analysis, providing clear insights into the chatbot's performance. A pilot test will be conducted with a small group of participants to ensure the clarity and reliability of the questionnaire. Based on feedback, the instrument will be refined before full-scale data collection. The final questionnaires will be distributed online, allowing participants to provide feedback after or during their interaction with the chatbot. The data collected will be analyzed using statistical methods to evaluate key metrics such as user satisfaction scores, response times, and usability ratings. Ethical considerations include obtaining informed consent from participants, maintaining confidentiality, and ensuring anonymity. This ensures that participants' rights are respected and that the data collection process aligns with ethical research standards.	
	COLLEGE OF ENGINEERING, COMPUTER STUDIES AND ARCHITECHTURE	53

Data Gathering Procedure

The data gathering procedure is a key part of the research methodology, ensuring the systematic collection of relevant data to evaluate the customizable chatbot plugin's effectiveness. This study will focus on collecting quantitative data using structured questionnaires.

Preparation and Recruitment. Before data collection begins, the research instruments will be finalized based on feedback from pilot testing. Necessary materials, including online survey platforms, will be prepared and tested for functionality to ensure a smooth data collection process. Participants will be recruited through departmental announcements, online forums, and direct email outreach. Collaboration with department heads will help ensure engagement and encourage participation from end-users, developers, and business stakeholders.

Data Collection Steps. Data collection will take place over a two-week period, during which participants will complete the structured online questionnaire after interacting with the chatbot. The questionnaires will be designed to collect quantitative responses that assess user satisfaction, feature usability, and system performance.

Data Management. Participants will be provided with an informed consent form explaining the study's purpose, procedures, and their rights. Their confidentiality and anonymity will be strictly maintained throughout the process. Responses will be anonymized during analysis, and participants will be informed of their right to withdraw from the study at any point without negative consequences.

The collected data will be stored securely in a password-protected database accessible only to the research team. Statistical analysis software (e.g., SPSS or R) will be used to process the data, generating insights through descriptive and inferential statistics.

Multiple Constraints Analysis

Criteria	Solution 1: Fully Custom In-House Development	Solution 2: Use Pre-Built AI Framework	Solution 3: Outsource Development to a Third Party	Recommended Solution
Financial	High cost due to resource allocation and tools	Moderate cost, with subscription or one-time fees	High cost due to external expertise	Solution 2: Offers affordability while providing advanced capabilities, aligning with academic constraints and budget limits.
Time	Long development cycle	Shorter due to existing framework availability	Medium, depends on third-party timelines	Solution 2: Faster implementation allows timely completion of thesis milestones and aligns with the research timeline.
Sampling	Requires significant effort to create test cases	Frameworks often have built-in support for user testing	Relies on third-party testing, may limit control	Solution 2: Built-in tools simplify evaluation processes, saving time and resources during the experimental phase of the thesis.

Technical Constraints	High customization but requires advanced skills	Limited by framework features, but still highly adaptable	High reliance on third-party expertise	Solution 2: Balances the need for technical feasibility and innovation while meeting thesis objectives without overcomplicating implementation.
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Summary of analysis:

The analysis of the three solutions for developing a customizable chatbot plugin fully custom in-house development, using a pre-built AI framework, and outsourcing development highlighted significant differences in their suitability for the thesis. Fully custom development offers the highest customization but is resource-intensive, requiring substantial time, technical expertise, and financial investment, which exceeds the typical academic scope. Outsourcing provides access to expert developers but is costly and limits control over the final product. The pre-built AI framework emerged as the optimal choice, balancing cost, time efficiency, and technical feasibility. It offers built-in tools for testing and evaluation, significantly reducing the workload for sampling and validation while meeting thesis deadlines. Furthermore, it provides sufficient adaptability to align with research goals without the complexity of custom development. This makes it a practical and efficient solution, ensuring the chatbot meets the constraints and objectives of the thesis effectively.

The data gathering procedure is a critical component of the research methodology, as it ensures the systematic collection of relevant information needed to evaluate the customizable

chatbot plugin's effectiveness. Data collection will be conducted using structured questionnaires and semi-structured interviews. This dual approach will provide comprehensive insights into user experiences and management perspectives, aligning with the study's objectives.

Preparation and Recruitment. Prior to data collection, the research instruments will be finalized based on feedback from pilot testing. All necessary materials, including online survey platforms and video conferencing tools for interviews, will be prepared and tested for functionality. Participants will be recruited through departmental announcements and direct outreach via email. Collaboration with department heads will facilitate engagement and encourage participation from both chatbot managers and general users.

Data Collection Steps. Data collection will take place over a two-week period, during which participants will complete the online survey after interacting with the chatbot. Interviews with the chatbot managers will be scheduled at convenient times via video conferencing, each lasting approximately 30-45 minutes.

Data Management. All participants will be provided with an informed consent form detailing the study's purpose, procedures, and their rights. Confidentiality and anonymity will be maintained throughout the process, with all responses anonymized in the analysis. Participants will also be informed of their right to withdraw from the study at any time without any negative consequences.

Collected data will be organized in a secure database, with access limited to the research team. Statistical analysis software (e.g., SPSS or R) will be utilized for quantitative data analysis, while qualitative data from interviews and observations will be analysed thematically.

Development Phases

The development of the customizable chatbot will follow a structured approach with distinct phases. First, during the Requirements Gathering phase (1-2 weeks), the researchers will conduct interviews and surveys with potential users, including the marketing department and its users, to identify their needs and desired functionalities. The delivery for this phase will be a comprehensive requirements specification document. Next, in the Design phase (3-4 weeks), the researchers will create wireframes and prototypes to visualize the user interface and interactions, resulting in high-fidelity mockups and design documentation.

The Implementation phase, which is expected to take 1-2 months, will involve developing the chatbot using programming languages such as Python and TypeScript, integrating key functionalities, and connecting it with backend systems. The outcome will be a fully functional chatbot plugin. Following implementation, the Testing phase (timeframe to be determined) will focus on unit testing, integration testing, and user acceptance testing (UAT) within three departments to ensure functionality and usability, with testing reports and bug fixes as deliverables.

The Deployment phase (timeframe to be recommended) will involve launching the chatbot within the targeted departments, ensuring proper configuration and accessibility, with the deliverable being the deployed chatbot plugin. Finally, the Maintenance phase will be ongoing, providing support, gathering user feedback, and implementing updates based on user experiences, with regular maintenance reports and update logs as key outputs.

Tools and Technologies. The development of the customizable chatbot will leverage a variety of tools and technologies to ensure efficient and scalable performance. For backend development, Python FastAPI will be utilized due to its high performance and ease of integration with other systems, while TypeScript and React will power the frontend, providing flexibility and component reusability. MySQL will serve as the database management system, ensuring reliable data storage and retrieval. These technologies were selected based on their compatibility with the project's requirements and the team's expertise, ensuring smooth development and deployment processes.

Team Roles and Responsibilities. The project team is structured to ensure clear roles and responsibilities throughout the development of the customizable chatbot. The first member is the team leader and main programmer, which is responsible for overseeing the entire development process, including coding, and integrating key functionalities. The second team member will act as the UI/UX designer, tasked with creating design mockups and ensuring that the user interface is intuitive and user-friendly. Lastly, the third member will serve as the documentation specialist, responsible for maintaining accurate and thorough documentation at every stage of development, ensuring all processes and decisions are well-documented.

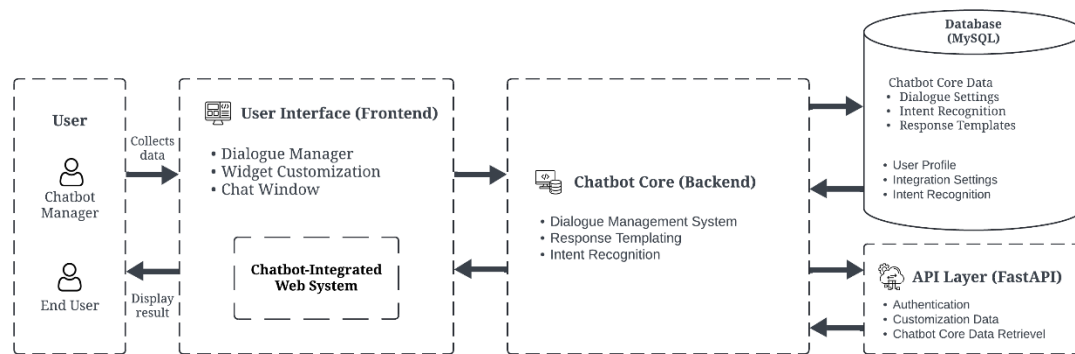
Feedback and Iteration. User feedback will be solicited at various stages, particularly during testing and after deployment. This feedback will be analyzed and used to inform subsequent iterations, enhancing the chatbot's functionality and overall user experience.

System Architecture

The system architecture of the customizable chatbot plugin is crafted to enable smooth interactions between users and backend services, ensuring both responsiveness and reliability. A well-structured architecture is essential for guaranteeing that all components operate cohesively to provide a functional and efficient chatbot experience. This section details the core components, technologies, and the interactions that underpin the system.

Architectural Design

Figure 2. Component Diagram



The component diagram outlines the architecture of a chatbot system, highlighting the interaction between frontend and backend components. The frontend features a Dialogue Manager for conversation flow, Widget Customization for personalizing the interface, and a Chat Window for end-user interaction, all integrated within the Chatbot Integrated Web System. There are two user roles: the End User, who engages with the chatbot, and the Chatbot Manager, who optimizes system performance. On the backend, the Chatbot Core includes a Dialogue Management System, Response Templating, and Intent Recognition. The API Layer, built with FastAPI, facilitates secure communication between the frontend and backend, managing authentication and customization data. The MySQL Database stores essential information such as user profiles and

integration settings. Together, these components create a cohesive chatbot system that enhances user experience.

Technology Stack

Frontend Technologies: The frontend will be built using TypeScript React for a dynamic user interface, complemented by Tailwind CSS frameworks for quick styling.

Backend Technologies: The backend will be developed using FastAPI to build efficient services, with Python powering the server-side logic. Sci-kit will be utilized for handling NLP and AI tasks, specifically for text classification.

Database: MySQL will be used for data storage, chosen for its reliability and ease of integration with the chosen backend framework.

Deployment Technologies: The chatbot will be deployed on the servers managed by the Information and Communication Technology Department of Lyceum of the Philippines Cavite, ensuring reliable access and system maintenance. The Communications and Public Affairs Department will be the primary users of the system, utilizing it to engage with potential university customers and facilitate seamless interaction and communication.

Data Flow

The data flow through the chatbot system begins when an End User accesses the chatbot via the Chat Window on the frontend, where they input a message or query. Upon submission, the

Dialogue Manager captures the user input and sends it to the API Layer, built with FastAPI, which serves as a secure bridge between the frontend and backend services. The API Layer handles authentication and retrieves relevant customization data before forwarding the input to the Chatbot Core. Within the Chatbot Core, the Dialogue Management System processes the input to determine its context and intent through natural language processing techniques. The Intent Recognition component identifies the user's intent, allowing the Response Templating component to generate an appropriate reply, often pulling information from the MySQL Database for personalization. This generated response is sent back to the API Layer, which forwards it to the Dialogue Manager, updating the Chat Window with the chatbot's reply for the End User to see. Meanwhile, the Chatbot Manager can interact with the system to optimize performance by accessing the Dialogue Manager and Widget Customization features, as well as querying the MySQL Database for insights into user profiles and conversation history. This flow ensures a seamless interaction between the user interface and backend services, providing relevant and personalized responses while supporting continuous improvement in the chatbot's performance.

Data Analysis

Data analysis is a crucial part of this study as it will help assess how well the customizable chatbot works and provide insights into user satisfaction and interactions. By analyzing the data collected, the study aims to evaluate the chatbot's performance and identify areas for improvement.

Data Collection Methods. The data will be gathered from three main sources: user interaction logs from the chatbot, surveys completed by users after their interactions, and feedback sessions with the department involved in testing. The data collected will include both quantitative information, such as response times and user engagement rates, and qualitative feedback, such as user satisfaction and suggestions for improvement. However, the focus of this study will be on analyzing the quantitative data to assess the chatbot's effectiveness through measurable metrics.

Before analysis, the data will be cleaned to remove any duplicates, irrelevant entries, or inconsistencies, ensuring that the dataset is reliable. The raw data will then be organized into structured formats suitable for statistical analysis and visualization. This will enable the researchers to apply appropriate statistical methods and draw meaningful conclusions from the data.

Analytical Techniques. The analysis will primarily focus on one department and its users. Descriptive analysis will be used to calculate summary statistics, such as averages (mean), most frequent values (mode), and middle values (median), to provide an overview of user interactions and satisfaction with the chatbot. To explore differences between user groups within the department, inferential analysis will be performed using statistical tests like t-tests or chi-square tests. These tests will help determine if there are significant differences in satisfaction levels between different groups of users. The goal is to identify factors that influence user engagement and satisfaction with the chatbot.

For analyzing user feedback, statistical methods will be employed to identify patterns and trends that can help improve the chatbot's features. These methods will allow the researchers to assess how various user characteristics, such as experience level or familiarity with chatbot technology, impact their overall satisfaction and engagement with the system.

Tools for Data Analysis. Basic statistical calculations and visualizations will be done using Excel, while more advanced analysis and graphical representations will be carried out using Python libraries like Pandas and Matplotlib. These tools will help process the data efficiently and generate clear, interpretable results that can guide the development of the chatbot.

Interpretation and Reporting. The results of the analysis will be interpreted to understand user engagement levels, satisfaction with the chatbot's performance, and how user characteristics relate to their feedback. These insights will be used to guide future improvements in the chatbot's design and functionality. The findings will be presented through written explanations and visual aids, such as charts and graphs, to make the results clear and accessible. A detailed report summarizing the key findings, conclusions, and recommendations for improving the chatbot will be created to inform future iterations and development of the system.

Ethical Considerations

In research involving human participants, ethical considerations are crucial. In this section, it will outline how the study will protect participants' rights, maintain confidentiality and guarantee responsibility in data usage.

In terms of participant's involvement and how their data will be used, they will be informed about the study's purpose before agreeing to participate in the study. Participants will be given

consent forms, displaying the nature of the study. This ensures participants can withdraw at any time without consequence. Additionally, they will have the right to access the data about them, enabling them to also request corrections if there are any inaccuracies found.

All data collected will be anonymized to maintain participant confidentiality. This ensures that no information related to identification is linked to the data. Both digital and physical data will also be securely stored in a protected environment, making sure that digital data is encrypted and stored in secure servers and cloud systems while physical data is put in a safeguarded place by the researchers. This limits access to only authorized personnels.

Collected Data will solely be used for research purposes related to development and improving the chatbot system. Data will be kept for the study, after which it will be safely disposed of to avoid unauthorized access. The study is expected to be very low-risk and if there are any potential discomfort, it will be addressed by proper assistance and communication.

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